

CANDIDATE
NAME

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CENTRE
NUMBER

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NUMBER

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CHEMISTRY

Grade 12

Paper 2

May 2014

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet
 Electronic calculator

12CHEM/02

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

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1	
2	
3	
4	
5	
6	
7	
8	
9	
Total	

This document consists of **15** printed pages and **1** blank page.

1 Propylamine, $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$, can be made by reacting 1-bromopropane with ammonia.

- (a) (i) Write a balanced chemical equation for the reaction of 1-bromopropane with ammonia, giving propylamine as one of the products.

..... [1]

- (ii) Name the mechanism of this reaction.

..... [1]

- (iii) Explain the mechanism of this reaction in the space below. Use curly arrows to show the movement of electron pairs. Label any relevant dipoles.

[3]

(b) Propylamine can also be made by the reduction of a suitable nitrile.

- (i) Name the nitrile that you would use.

..... [1]

- (ii) Name the reducing agent that you would use.

..... [1]

- (iii) Give one essential condition for this reaction.

..... [1]

(c) (i) Write a balanced chemical or ionic equation to show ammonia acting as a base.

..... [1]

- (ii) Propylamine is also a base, but it is a stronger base than ammonia.

Explain why.

.....

..... [1]

[Total: 10]

- 2 The element sulfur has several isotopes, some of which, such as ^{32}S , ^{33}S and ^{34}S , are stable. Other isotopes such as ^{35}S , are radioactive. Sulfur atoms can form sulfide ions, S^{2-} .

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- (a) Explain the meaning of the term *isotopes*.

.....
..... [1]

- (b) Complete the table to show the numbers of protons, neutrons, and electrons in a ^{32}S atom, a ^{33}S atom, and a $^{33}\text{S}^{2-}$ ion.

species	^{32}S	^{33}S	$^{33}\text{S}^{2-}$
protons			
neutrons			
electrons			

[3]

- (c) Sodium and sulfur will react together to form sodium sulfide. Draw a 'dots and crosses' diagram to show the bonding in sodium sulfide.

Name the type of bonding involved.

type of bonding [2]

- (d) ^{35}S undergoes β decay. Each ^{35}S atom emits one β particle.

- (i) What other name is commonly given to a β particle?

..... [1]

- (ii) Write a nuclear equation for the β decay of one ^{35}S atom.

..... [1]

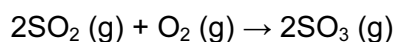
- (e) A sample of sulfur produced by a research establishment was analysed using a mass spectrometer. It was found to consist of: 94.94% ^{32}S , 0.77% ^{33}S , 4.27% ^{34}S , and 0.02% ^{35}S .

Calculate the relative atomic mass of the sample of sulfur. Give your answer correct to **four** significant figures.

answer [2]

[Total: 10]

- 3 Sulfuric acid is manufactured industrially by a sequence of reactions, the most important of which is called the Contact process. In the Contact process sulfur dioxide reacts with oxygen to produce sulfur trioxide:



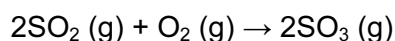
Use this data to answer some of the questions that follow.

	SO_2	SO_3	O_2
standard enthalpy of formation ($\Delta H_f^\ominus$) in kJ mol^{-1}	-297	-395	
standard entropy ($S^\ominus$) in $\text{J K}^{-1} \text{mol}^{-1}$	248	256	205

- (a) What is meant by the term *standard enthalpy of formation*?

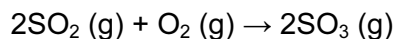
..... [2]

- (b) Calculate the standard **enthalpy** change, $\Delta H^\ominus$, of the reaction



..... [2]

- (c) (i) Calculate the standard **entropy** change, ΔS° , of the reaction



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..... [2]

- (ii) Explain, by reference to the equation $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$, the **sign** (positive or negative) of your answer for the standard **entropy** change of the reaction.

..... [1]

- (d) Use your answers to parts (b) and (c)(i) to calculate the maximum temperature at which the reaction $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$ will proceed spontaneously. Assume ΔH and ΔS stay constant as temperature increases.

maximum temperatureunits [4]

[Total: 11]

4 Ammonia is a compound of nitrogen and hydrogen. Its formula is NH_3 .

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(a) (i) Draw an ammonia molecule, showing its shape in three dimensions.

[1]

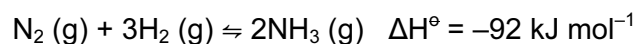
(ii) What is the value of the H-N-H bond angle in ammonia?

[1]

(iii) Explain the shape of the ammonia molecule, include in your answer an explanation for the value of the H-N-H bond angle.

[2]

(b) Ammonia can be made by direct synthesis if a mixture of nitrogen and hydrogen gases is passed over a suitable catalyst. The equation is



A chemist puts 6.0 g (3.0 mol) of hydrogen and 28.0 g (1.0 mol) of nitrogen into a sealed vessel of volume 1.0 dm^3 . The vessel contains a powdered iron catalyst. The vessel and its contents were maintained at a temperature of 550°C and left until a dynamic equilibrium was established. The chemist then analysed the contents of the vessel, which was found to contain 6.8 g of ammonia.

(i) What are the characteristics of a system when **dynamic equilibrium** is established?

[3]

(ii) How many moles of ammonia were present at equilibrium?

[1]

- (iii) Write an expression for K_c for this reaction

$$K_c =$$

For
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Use

[1]

- (iv) Calculate the numerical value of K_c at 550 °C. Give your answer correct to **two** significant figures.

$$K_c = \dots\dots\dots [3]$$

- (v) How would you expect the numerical value of K_c to change if the chemist had used a temperature of 700 °C?

Explain your answer.

.....

.....

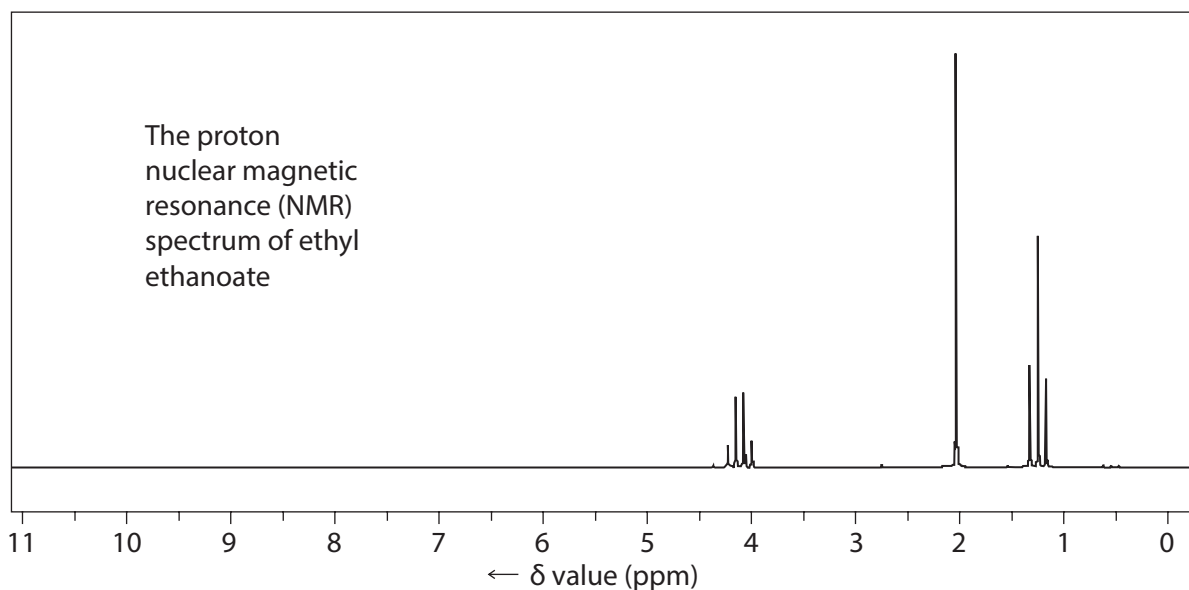
..... [2]

[Total: 14]

5 Ethyl ethanoate is an ester, its structural formula is $\text{CH}_3\text{COOC}_2\text{H}_5$.

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- (a) The proton nuclear magnetic resonance (^1H NMR) spectrum of ethyl ethanoate shows three absorptions with δ values 4.1 ppm, 2.0 ppm, and 1.3 ppm.



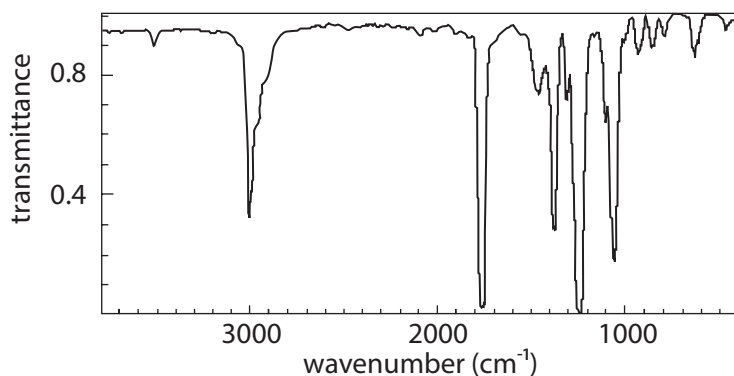
- (i) Explain why the absorption with a δ value of 4.1 ppm is split into four peaks.

[1]

- (ii) Identify the hydrogen atoms in the ethyl ethanoate molecules that are responsible for each of the three absorptions with δ values 4.1 ppm, 2.0 ppm, and 1.3 ppm.

[2]

- (b) The infra-red (IR) spectrum of ethyl ethanoate is shown below. It has a prominent absorption at approximately 1750 cm^{-1} .



The infra-red (IR) spectrum of ethyl ethanoate

- (i) Explain why molecules of ethyl ethanoate are able to absorb infra-red radiation.

.....
 [2]

- (ii) Identify the structural feature in each ethyl ethanoate molecule that is responsible for the absorption at approximately 1750 cm^{-1} .

..... [1]

- (c) Ethyl ethanoate can be hydrolysed by heating under reflux with dilute aqueous sulfuric acid or by boiling under reflux with dilute aqueous sodium hydroxide.

- (i) Write a balanced chemical equation for each of these hydrolysis reactions.

with sulfuric acid

with sodium hydroxide
 [2]

- (ii) Choose one of the hydrolysis reactions in (c)(i) and **name** both products of the reaction.

..... [1]

- (iii) A sample of ethyl ethanoate is heated under reflux with dilute aqueous sulfuric acid for one hour. A second sample of ethyl ethanoate is heated under reflux with dilute aqueous sodium hydroxide for one hour.

Explain why these two hydrolysis reactions produce different percentage yields of the products.

.....

 [2]

[Total: 11]

- 6 Group 14 of the Periodic Table consists of carbon, silicon, germanium, tin and lead. Carbon can exist in three different allotropic forms, diamond, graphite, and buckminsterfullerene.

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- (a) The melting point of graphite is much higher than the melting point of lead.
Explain why this is, with reference to the structure and bonding of the two substances.

.....
.....
.....
..... [3]

- (b) Buckminsterfullerene has a simple molecular structure, it also has a much lower melting point than graphite.

Give the molecular formula of buckminsterfullerene and name the forces that hold buckminsterfullerene molecules together in the solid state.

.....
..... [2]

- (c) Describe and explain the trend in first ionisation energy as group 14 is descended from carbon to lead.

.....
.....
.....
..... [3]

(d) The elements in group 14 form oxides in two different oxidation states.

- (i) Give the formulae of the two oxides of carbon and the two oxides of lead, and give the oxidation state of the group fourteen element in each compound.

oxide of carbon oxidation state of carbon in this oxide

oxide of carbon oxidation state of carbon in this oxide

oxide of lead oxidation state of lead in this oxide

oxide of lead oxidation state of lead in this oxide [2]

- (ii) Describe the relative stability of the two oxides of carbon, illustrating your answer with a chemical equation for a reaction in which the more stable oxide is formed from the less stable oxide.

.....

..... [2]

[Total: 12]

7 Compound **Q** is a compound of carbon, hydrogen and oxygen only. **Q** has a simple molecular structure.

- (a) A sample of **Q** of mass 0.2250 g is vapourised and produces 4155 cm³ of gas at a temperature of 400 K and a pressure of 2,000 Pa.

Use the ideal gas equation to calculate the number of moles of **Q** in the 0.2250 g sample.

.....mol [3]

- (b) (i) Further analysis of **Q** shows that it consists of 40.00% carbon, 6.67% hydrogen, and 53.33% oxygen by mass.

Calculate the empirical formula of **Q**.

empirical formula [2]

- (ii) Calculate the M_r value of **Q** and hence its molecular formula.

molecular formula [1]

- (c) **Q** is a secondary alcohol.

What would be **observed** if a sample of **Q** was heated with acidified aqueous potassium dichromate?

..... [1]

[Total: 7]

- 8 Copper(II) sulfate, CuSO_4 , dissolves in water to produce a pale blue solution. The blue colour of this aqueous solution is due to the presence of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ complex ions.

- (a) Draw a $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ complex ion, showing its three dimensional shape. Label one of the bond angles at the central Cu^{2+} ion. The value of this bond angle should be shown.

Cu

[2]

- (b) If sodium hydroxide solution is added dropwise to an aqueous solution of copper(II) sulfate, a chemical reaction occurs.

- (i) What observations can be made during this chemical reaction?

..... [1]

- (ii) Write the balanced chemical equation for this chemical reaction.

..... [1]

- (iii) The chemical reaction between sodium hydroxide solution and copper(II) sulfate solution can be described as an **acid-base** reaction.

Explain this statement as fully as you can.

.....

 [3]

(c) If an excess of concentrated aqueous hydrochloric acid is added to an aqueous solution of copper(II) sulfate a chemical reaction occurs.

(i) What observations can be made during this chemical reaction?

..... [1]

(ii) Complete the balanced chemical equation for this chemical reaction, giving in full the formulae of any complex ions involved.



(iii) What is the name of this type of chemical reaction?

..... [1]

(iv) During this reaction the Cu^{2+} ions undergo a change in coordination number.

What was the original coordination number of the Cu^{2+} ions, and what was it after the reaction?

original coordination number of the Cu^{2+} ions

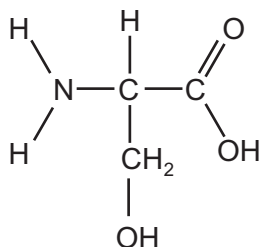
coordination number of the Cu^{2+} ions after the reaction [1]

(v) Why does this reaction result in a change of the coordination number of the Cu^{2+} ions?

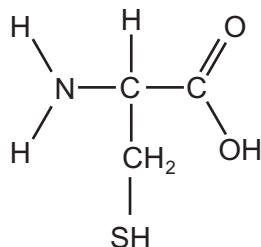
.....
..... [1]

[Total: 13]

9 Proteins are naturally occurring macromolecules made by the polymerisation of amino acids. Two of these amino acids are serine and cysteine.



serine



cysteine

(a) What is the systematic name of serine?

..... [1]

(b) Draw the structure of cysteine in the **zwitterion** form in the space below.

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[1]

(c) Two amino acid molecules may react together, forming a peptide link. The product of this reaction is called a dipeptide. A serine molecule and a cysteine molecule can react together to produce two different dipeptides.

Draw the displayed formulae of the two dipeptides in the space below, labelling the peptide link in one of your diagrams.

[3]

(d) Insulin is a hormone found in humans and other animals. Each insulin molecule consists of two chains of amino acids called the α -chain and the β -chain. The α -chain is joined to the β -chain by linkages that are formed between the R-groups of cysteine residues. The linkage is a common one, found in many other proteins.

(i) What is the R-group of a cysteine residue?

..... [1]

(ii) In the space below show one linkage formed between the R-groups of two cysteine residues.

[1]

(iii) Name the linkage you have drawn in (ii) [1]

(e) An insulin molecule, consisting of one α -chain joined to one β -chain, folds up into a three-dimensional shape.

(i) What is the name of the level of protein structure produced by this folding process?

..... [1]

(ii) What interaction between the R-groups of two serine residues could help to stabilise the folded insulin?

..... [1]

(f) A scientist investigates the structure of insulin by partially hydrolysing the α and β -chains and then analysing the short lengths of chain that are produced. One of the short lengths of chain, **X**, consists of seven amino acid residues joined together by peptide links. Only five different tripeptides can be obtained by further hydrolysis of **X**. These tripeptides are:

arg-glu-gly

tyr-phe-phe

gly-arg-glu

phe-phe-gly

phe-gly-arg

What is the amino acid sequence of **X**?

..... [2]

[Total: 12]

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